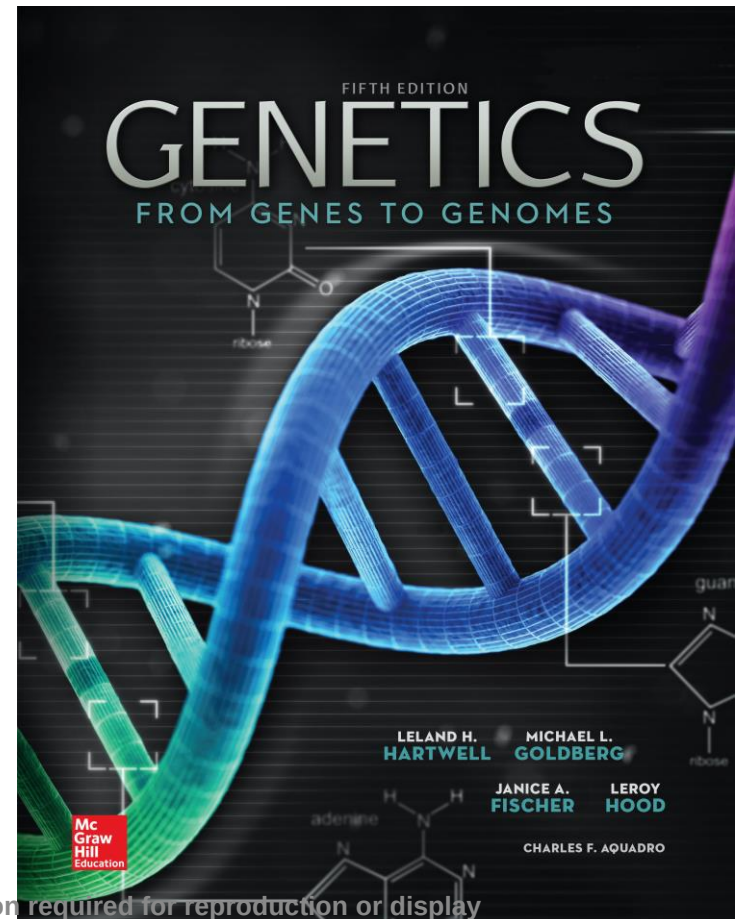


Genetics: From Genes to Genomes

Fifth Edition

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Mendel's Principles of Heredity



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CHAPTER OUTLINE

- 2.1 Background: The Historical Puzzle of Inheritance
- 2.2 Genetic Analysis According to Mendel
- 2.3 Mendelian Inheritance in Humans

2.1 Background: The Historical Puzzle of Inheritance

Learning Objectives:

- **Relate how Mendel's experimental approach is similar to the process of modern scientific inquiry.**
- **Describe how Mendel cross-fertilized and self-fertilized pea plants.**
- **Explain the importance of Mendel's inclusion of reciprocal crosses within his controlled breeding program.**
- **Predict the type of progeny produced by Mendel's crosses between pure-breeding plants with discrete, antagonistic traits.**

Background to Mendel's work: The historical puzzle of inheritance

Artificial selection was the first applied genetic technique

- Purposeful control of mating by choice of parents for the next generation
- Domestication of plants and animals was a key transition in human civilization





same for people
family picture showing differences and similarities
who is NOT related?
conclude traits are inherited but somewhat random -

Genetics before Mendel

- artificial selection in plant breeding, domestic animals
- traits tend to be passed on
 - ...but with many exceptions:
 - two children from the same parents may appear very different
 - children may appear more like grandparents or a more distant relative
 - even apparently pure breeds can produce unexpected variation
- there was no understanding of how traits could be passed on

Critical questions about selective breeding before Mendel's studies

The results of selective breeding were unpredictable.

In 1837, Abbot Cyril Napp proposed three basic questions:

- **What is inherited?**
- **How is it inherited?**
- **What is the role of chance in heredity?**



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Abbot Napp presided over the monastery where Mendel began his seminal genetic experiments in 1864 .

Two misleading theories of inheritance at the time of Mendel's studies

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The homunculus: Inherited features of offspring are contributed mainly by only one parent (*they thought that the human being comes only from the sperm and the egg is only needed for nourishment*)

Blended inheritance: Parental traits become mixed and changed in the offspring



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The homunculus

Gregor Mendel discovered the basic principles of genetics

Mendel was the first scientist to combine data collection, analysis, and theory to understand heredity.

He inferred genetic laws about the appearance and disappearance of traits in different generations.

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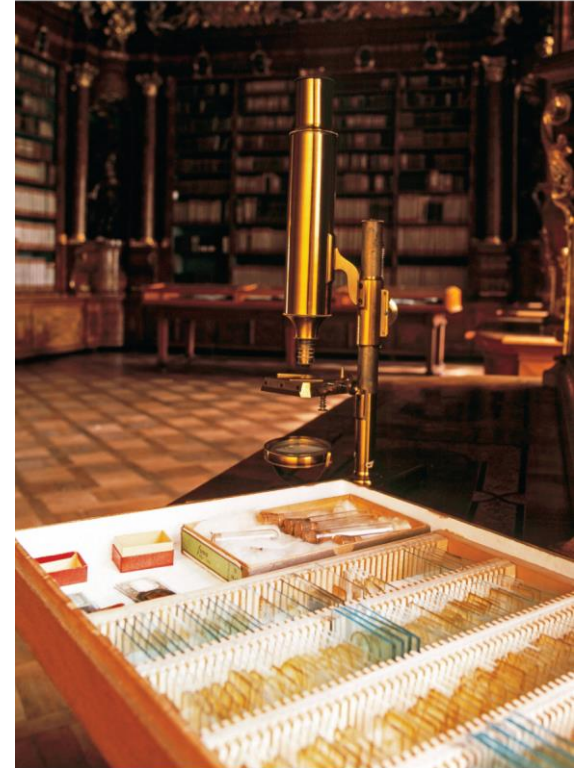
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Mendel studied the inheritance of alternative traits in pea plants

Mendel inferred laws of genetics that allowed predictions about which traits would appear, disappear, and then reappear.

Mendel's paper "Experiments in plant hybrids" was published in 1866 and became the cornerstone of modern genetics.

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Microscope used by Mendel

Four general themes of Mendel's work

- 1. Variation is widespread in nature and provides for continuously evolving diversity**
- 2. Observable variation is essential for following genes from one generation to another**
- 3. Variation is inherited by genetic laws, which can explain why like produces like and unlike (e.g. Fig 2.3)**
- 4. Mendel's laws apply to all sexually reproducing organisms**

Genetic variation exists even within dog breeds

Mendel's laws explain why two black Labradors could have a litter of black, brown, and golden puppies.



© Saudjie Cross Sino/Weathertop Labradors

Keys to the success of Mendel's experiments

Pure-breeding lines of peas (*Pisum sativum*)

- Breeding could be done by cross-fertilization or selfing
- Large numbers of progeny produced within a short time
- Traits remained constant in crosses within a line

Inheritance of alternative forms of traits

- Antagonistic pairs of "either-or" traits: e.g. purple or white, yellow or green

Brilliant experimentalist

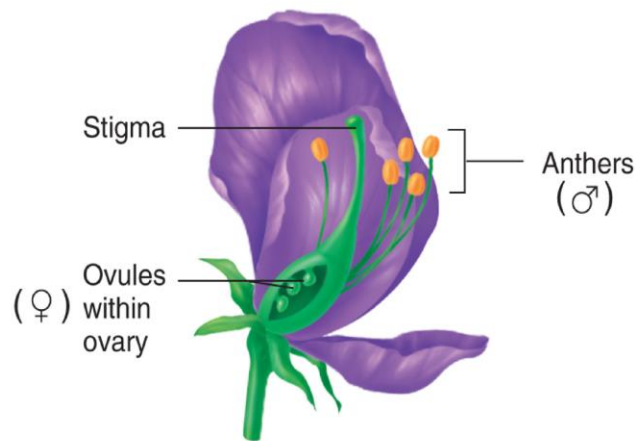
- Carefully planned and carried-out experiments
- Controlled the plant breeding, including reciprocal crosses
- Analyzed results mathematically

Mendel's experimental organism: The garden pea

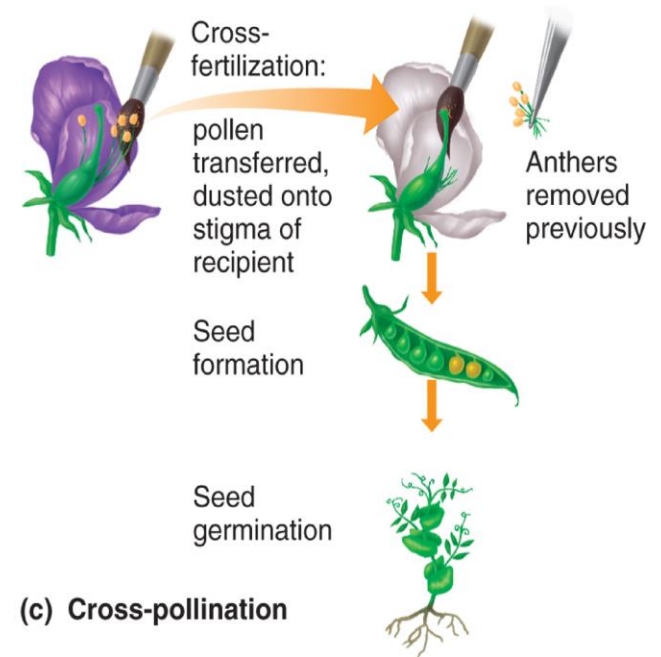
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(a) *Pisum sativum*



(b) Pea flower anatomy



(c) Cross-pollination

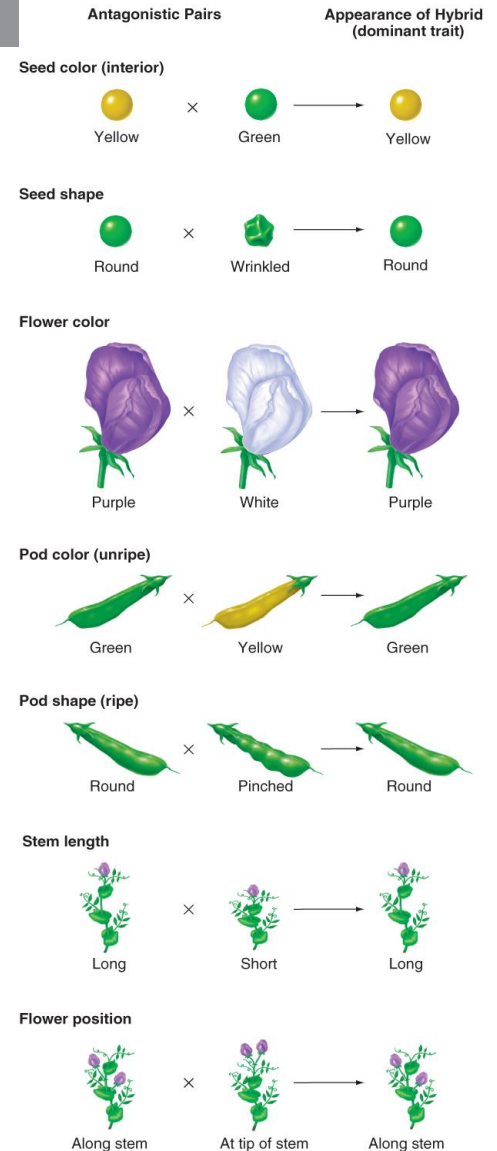
Mendel studied seven antagonistic pairs of traits in peas

Three antagonistic pairs of traits are shown at right.

The anthers contain the pollens

Note that each hybrid resembles only one of the parents.

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2.2 Genetic Analysis According to Mendel

Learning Objectives:

- **Explain Mendel's law of segregation and predict the phenotypic ratio in the F₂ of a monohybrid cross.**
- **Distinguish between a monohybrid cross and a testcross.**
- **Explain Mendel's law of independent assortment and provide evidence for it using a dihybrid cross.**
- **Interpret phenotypic ratios to infer how traits are inherited.**
- **Predict the genotypic and phenotypic ratios among progeny of complex multihybrid crosses using simple rules of probability.**
- **Cite the most common molecular explanations for dominant and recessive alleles.**

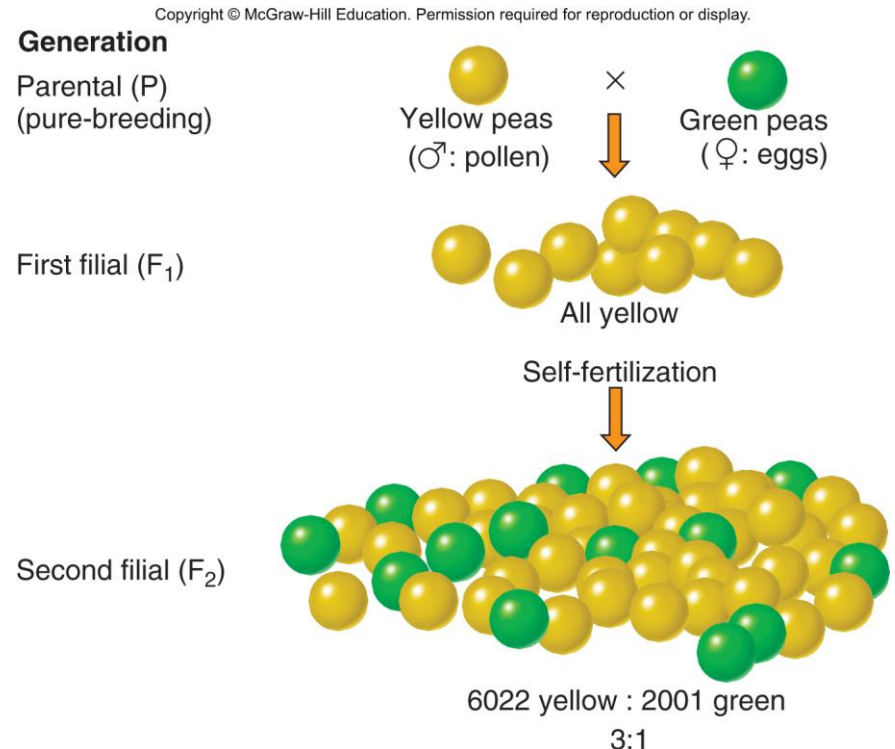
Monohybrid crosses revealed units of inheritance and the law of segregation

Mendel crossed pure-breeding lines that differed in only one trait, e.g. seed color

Examined phenotypes of F_1 progeny and F_2 progeny

- F_1 progeny have only one of the parental traits
- Both parental traits reappear in F_2 progeny in a 3:1 ratio

These results disproved the the blending hypothesis.

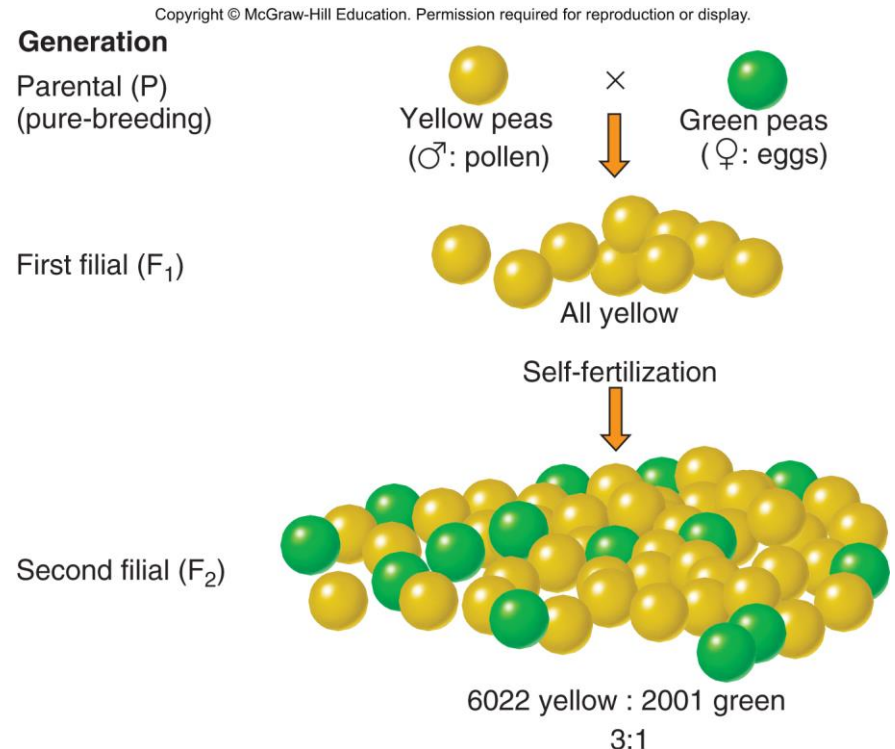


Monohybrid crosses revealed units of inheritance and the law of segregation

Mendel performed reciprocal crosses

- Pollen from yellow peas fertilized eggs from green peas (shown)
- Pollen from green peas fertilized eggs from yellow peas

The results from reciprocal crosses produced similar results, disproving the blending hypothesis.



Mendel proposed that each plant carries two copies of a unit of inheritance

Traits have two forms that can each breed true

- Trait that appears in F_1 progeny is the **dominant** form
- Trait that is hidden in the F_1 progeny is the **recessive** form

Units of inheritance are now known as "**genes**"

- Alternative forms of a single gene are "**alleles**"
- Individuals with two different alleles for a single trait are "**monohybrids**"

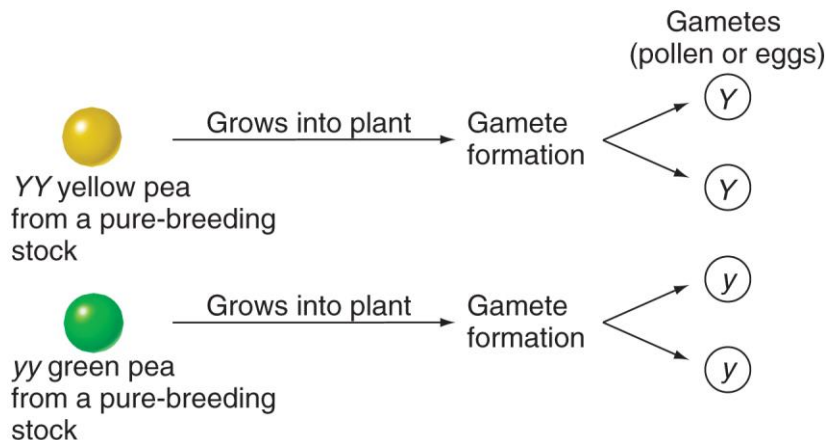
How do progeny inherit one unit from the maternal parent and the other unit from the paternal parent?

Mendel's law of segregation

The two alleles for each trait separate during gamete formation

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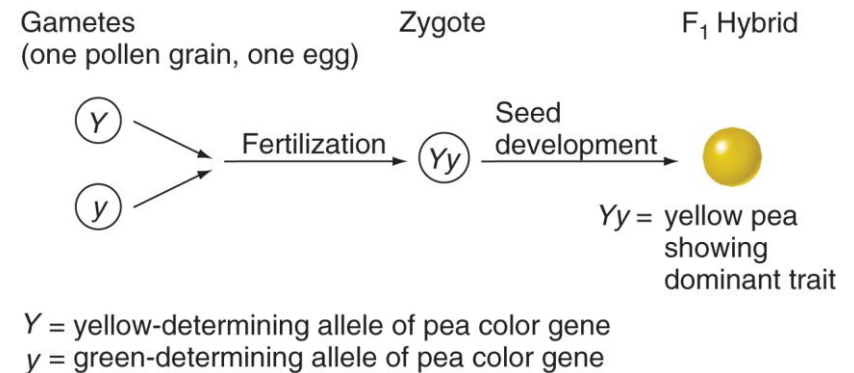
(a) The two alleles for each trait separate during gamete formation.



Two gametes, one from each parent, unite at random at fertilization

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(b) Two gametes, one from each parent, unite at random at fertilization.

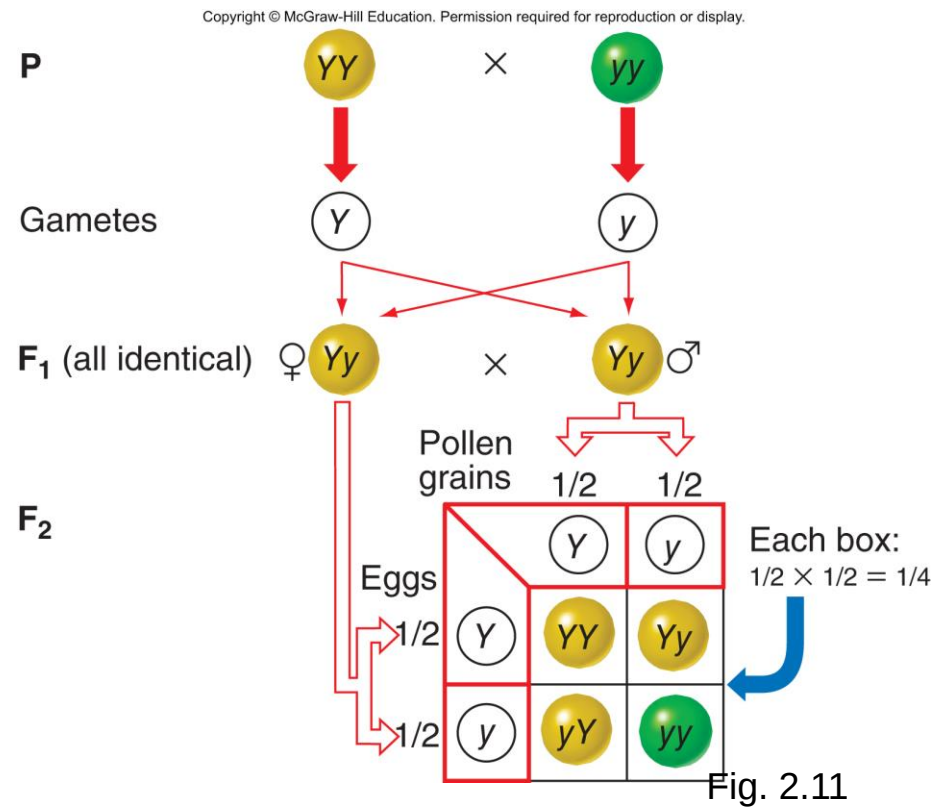


The Punnett square is a simple way to visualize the segregation and random union of alleles

Each F_1 hybrid produces two kinds of gametes in a 1:1 ratio

F_2 progeny

- 3:1 ratio of phenotypes
- 1/4 will breed true for the dominant trait
- 1/2 will be hybrids
- 1/4 will breed true for the recessive trait



Mendel's results and the Punnett square reflect the basic rules of probability

Product rule: probability of two independent events occurring together is the product of their individual probabilities

- What is the probability that event 1 AND event 2 will occur?

$$P(1 \text{ and } 2) = \text{probability of event 1} \times \text{probability of event 2}$$

Sum rule: probability of either of two mutually exclusive events occurring is the sum of their individual probabilities

- What is the probability that event 1 OR event 2 will occur?

$$P(1 \text{ or } 2) = \text{probability of event 1} + \text{probability of event 2}$$

Applying probability to Mendel's crosses

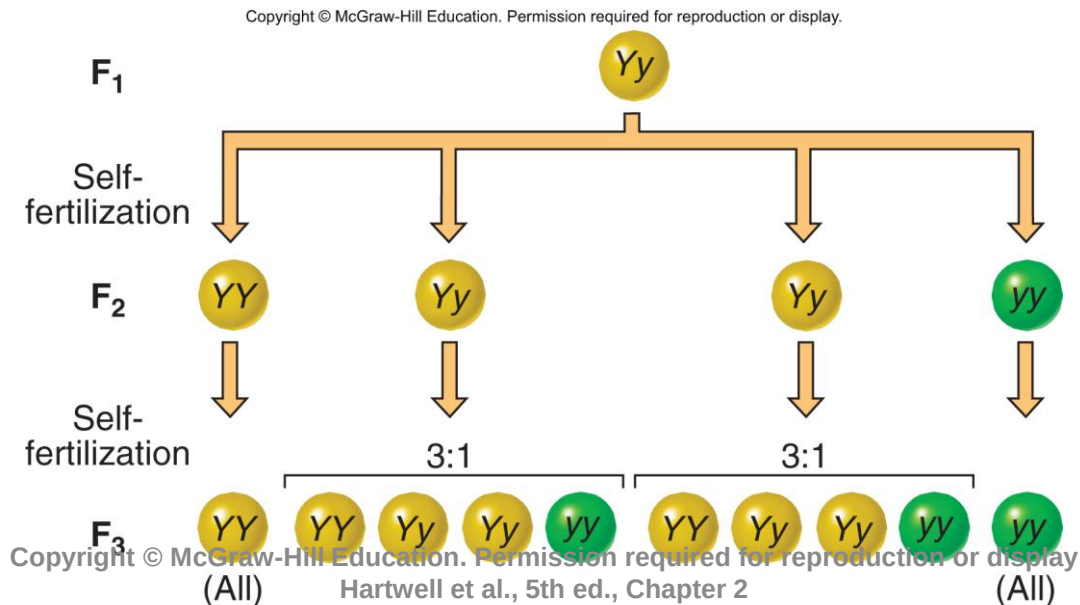
From a cross of $Yy \times Yy$ peas

- What is the chance of getting YY offspring?
 - Chance of Y pollen is $1/2$
 - Chance of Y ovule is $1/2$
 - Chance of Y pollen and Y ovule uniting is $1/2 \times 1/2 = 1/4$
- What is the chance of getting Yy offspring?
 - Chance of Y pollen uniting with y ovule is $1/2 \times 1/2 = 1/4$
 - Chance of y pollen uniting with Y ovule is $1/2 \times 1/2 = 1/4$
 - Chance of either event happening is $1/4 + 1/4 = 1/2$

Mendel did further crosses to verify the law of segregation

F₂ plants were selfed to produce F₃ progeny

- All of the green F₂ peas were pure breeding
- 1/3 of the yellow F₂ peas were pure breeding
- 2/3 of the yellow F₂ peas were hybrids



Definitions of commonly used terms

Phenotype is an observable characteristic (e.g. yellow or green pea seeds)

Genotype is a pair of alleles in an individual (e.g. YY or Yy)

Homozygote has two identical alleles (e.g. YY or yy)

Heterozygote has two different alleles (e.g. Yy)

- The heterozygous phenotype defines the dominant allele (e.g. Yy peas are yellow, so the yellow Y allele is dominant to the green y allele)
- A dominant allele with a dash represents an unknown genotype (e.g. Y– stands for either YY or Yy)

Genotype vs phenotype in homozygotes and heterozygotes

From a cross of $Yy \times Yy$ peas

Genotypes in F_2 progeny are in 1:2:1 ratio ($1/4 YY$, $1/2 Yy$, $1/4 yy$)

Phenotypes in F_2 progeny are in 3:1 ratio ($3/4$ yellow, $1/4$ green)

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Genotype for the Seed Color Gene	Phenotype
YY Homozygous dominant	 Yellow
Dominant allele ——— ——— Recessive allele Yy Heterozygous	 Yellow
yy Homozygous recessive	 Green

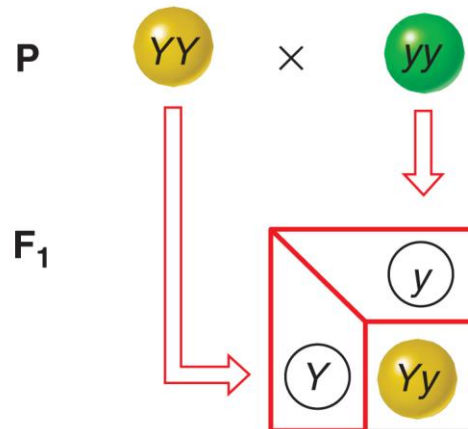
A testcross can reveal an unknown genotype

Is the genotype of an individual with a dominant phenotype (e.g. $Y-$) heterozygous (Yy) or homozygous (YY)?

- Solution: Testcross to homozygous recessive (yy) and examine progeny

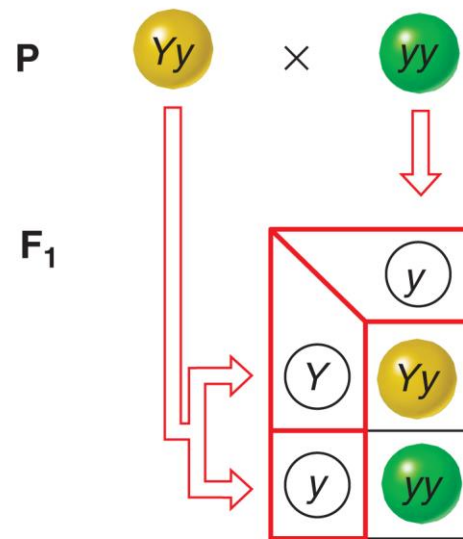
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Cross A



Offspring all yellow

Cross B



Offspring 1:1 yellow to green

Mendel's dihybrid crosses revealed the law of independent assortment

Mendel tested whether two genes in dihybrids would segregate independently

First, he crossed true-breeding yellow round peas with true-breeding green wrinkled peas to obtain dihybrid F_1 plants:

$$YY RR \times yy rr \rightarrow F_1 Yy Rr$$

Then, the dihybrid F_1 plants were selfed to obtain F_2 plants:

$$F_1 Yy Rr \times F_1 Yy Rr \rightarrow F_2$$

Mendel asked whether all the F_2 progeny would be **parental types** (yellow round and green wrinkled) or would some be **recombinant types** (yellow wrinkled and green round)?

A dihybrid cross produces parental types and recombinant types

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Type	Genotype	Phenotype	Number	Phenotypic Ratio
Parental	$Y- R-$	 yellow round	315	9/16
Recombinant	$yy R-$	 green round	108	3/16
Recombinant	$Y- rr$	 yellow wrinkled	101	3/16
Parental	$yy rr$	 green wrinkled	32	1/16

Ratio of yellow (dominant) to green (recessive) = 12:4 or 3:1

Ratio of round (dominant) to wrinkled (recessive) = 12:4 or 3:1

Fig. 2.15

Independent assortment in crosses of F_1 dihybrids produces a 9:3:3:1 phenotype ratio

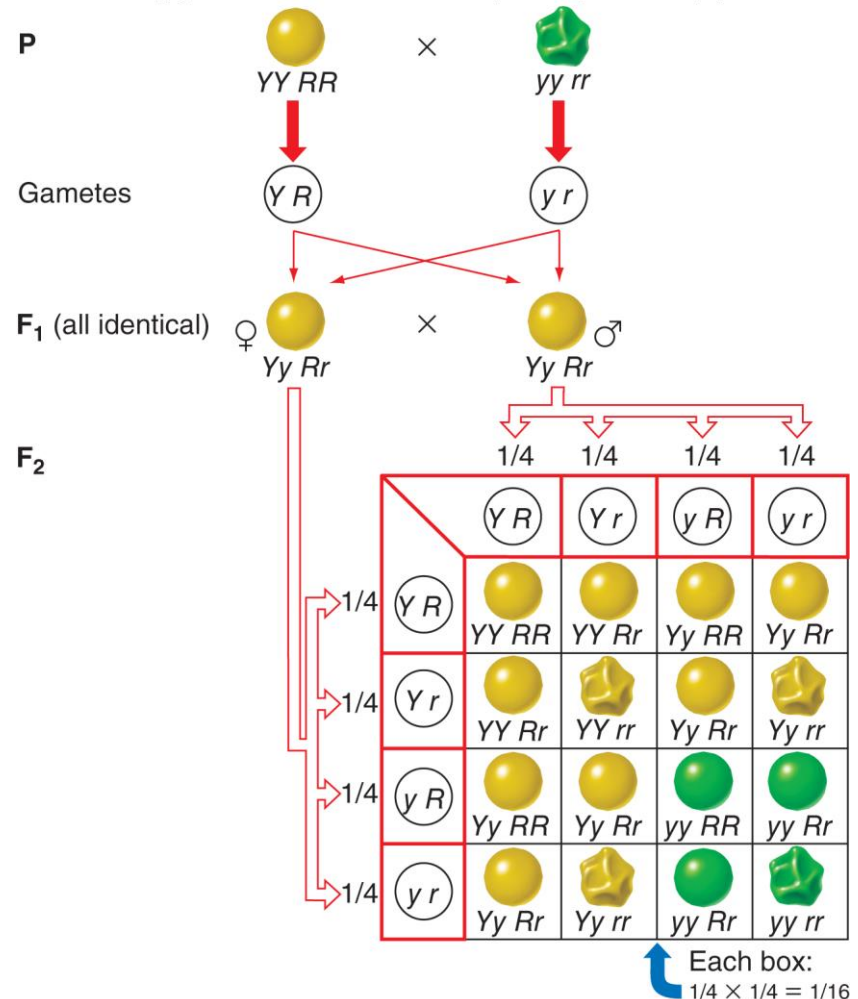
Each F_1 dihybrid produces four possible gametes in a 1:1:1:1 ratio

$$Yy Rr \rightarrow 1/4 YR, 1/4 Yr, 1/4 yR, 1/4 yr$$

Four phenotypic classes occurred in the F_2 progeny:

- Two are like parents
- Two are recombinant

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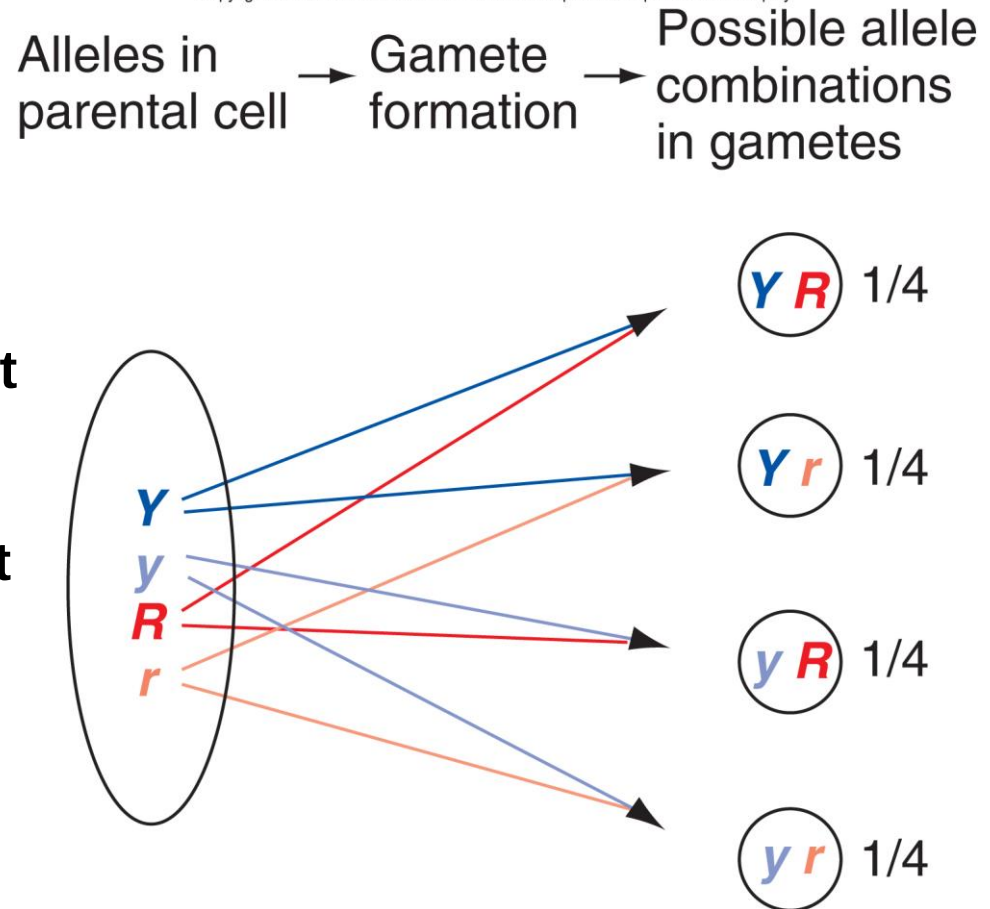


Mendel's law of independent assortment

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During gamete formation, different pairs of alleles segregate independently of each other

- ***Y* is just as likely to assort with *R* as it is with *r***
- ***y* is just as likely to assort with *R* as it is with *r***



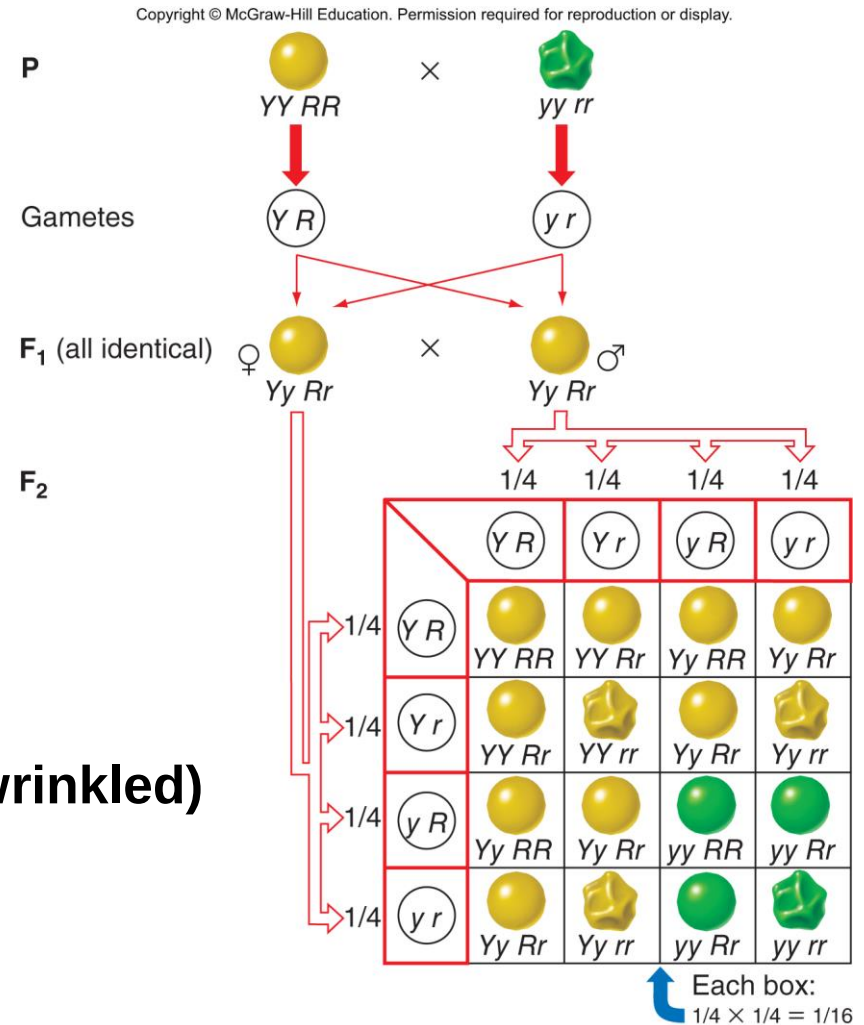
Using the product rule to calculate the ratios observed in the F₂ of a dihybrid cross

Each trait is inherited in a 3:1 ratio

- Ratio of yellow (dominant) to green (recessive) = 3:1
- Ratio of round (dominant) to wrinkled (recessive) = 3:1

Product rule

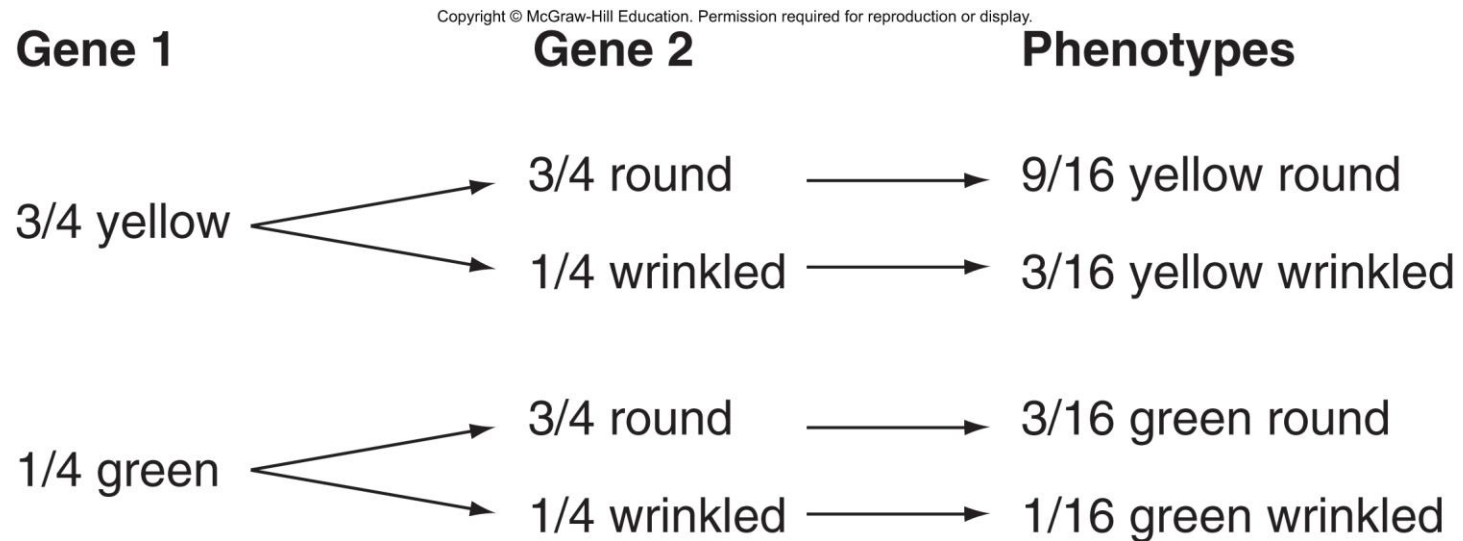
$$\begin{aligned}
 P(\text{yellow, wrinkled}) &= P(\text{yellow}) \times P(\text{wrinkled}) \\
 &= \frac{3}{4} \times \frac{1}{4} \\
 &= \frac{3}{16}
 \end{aligned}$$



Following crosses with branched-line diagrams

Progeny phenotypes for each gene are shown in different columns

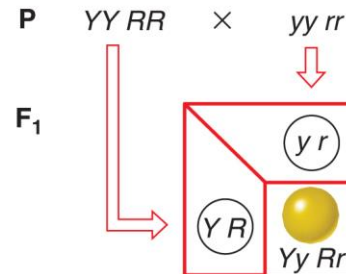
This gives the same ratios as seen in the Punnett square in Fig 2.15



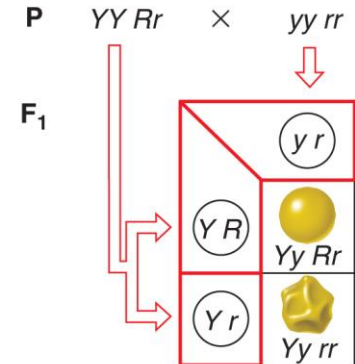
Testcrosses on dihybrids

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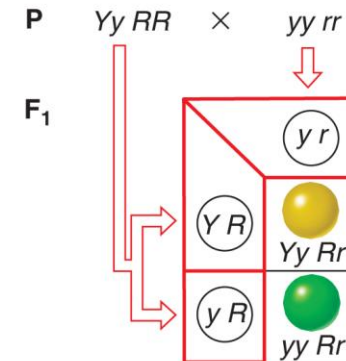
Cross A



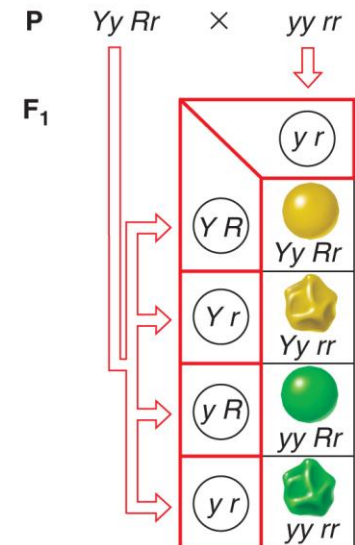
Cross B



Cross C



Cross D



Testcross dihybrids to individuals that are homozygous for both recessive traits

Mendel's laws can be used to predict offspring from complicated crosses

**The number of different gametes = 2^n
(n = # of different traits)**

- ***Aa Bb Cc Dd* $\rightarrow 2^4 = 16$ kinds of gametes**
- ***Aa Bb Cc Dd* $\times$ *Aa Bb Cc Dd* $\rightarrow 16 \times 16 = 256$ genotypes**
- **To do a Punnett square with this cross involving four genes, you would need 16 columns and 16 rows**
- **An easier way is to break down a multihybrid cross into independently assorting monohybrid crosses**

Predicting proportions of progeny from multihybrid crosses – example 1

Cross *Aa Bb Cc Dd* x *Aa Bb Cc Dd*

What proportion of progeny will be *AA bb Cc Dd*?

- *Aa* x *Aa* → 1/4 *AA*
- *Bb* x *Bb* → 1/4 *bb*
- *Cc* x *Cc* → 1/2 *Cc*
- *Dd* x *Dd* → 1/2 *Dd*

So, the expected proportion of *AA bb Cc Dd* progeny is:

$$1/4 \times 1/4 \times 1/2 \times 1/2 = 1/64$$

Predicting proportions of progeny from multihybrid crosses – example 2

Cross $Aa Bb Cc Dd \times Aa Bb Cc Dd$

How many progeny will show the dominant traits for A , C , and D and the recessive trait for B ?

- $Aa \times Aa \rightarrow 3/4 A-$
- $Bb \times Bb \rightarrow 1/4 bb$
- $Cc \times Cc \rightarrow 3/4 C-$
- $Dd \times Dd \rightarrow 3/4 D-$

So, expected proportion of $A- bb C- D-$ progeny is:

$$3/4 \times 1/4 \times 3/4 \times 3/4 = 27/256$$

The science of genetics began with the rediscovery of Mendel's work

Mendel published his monumental breakthrough in understanding heredity in 1866, but hardly anyone paid attention to his work!

In 1900, three scientists independently rediscovered and acknowledged Mendel's work

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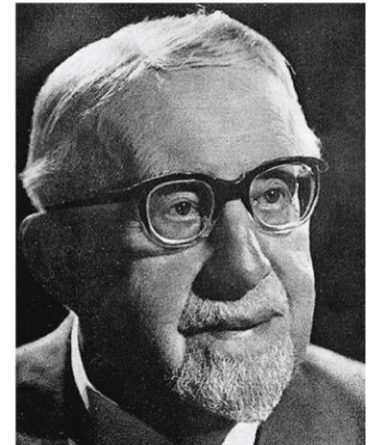
(a) Gregor Mendel



(b) Carl Correns



(c) Hugo de Vries



(d) Erich von Tschermak

(a): © Science Photo Library/Science Source, (b–d): © Mendelianum Institute, Moravian Museum

Molecular explanation for Mendel's pea shape

The R allele encodes Starch branching enzyme (Sbe1).

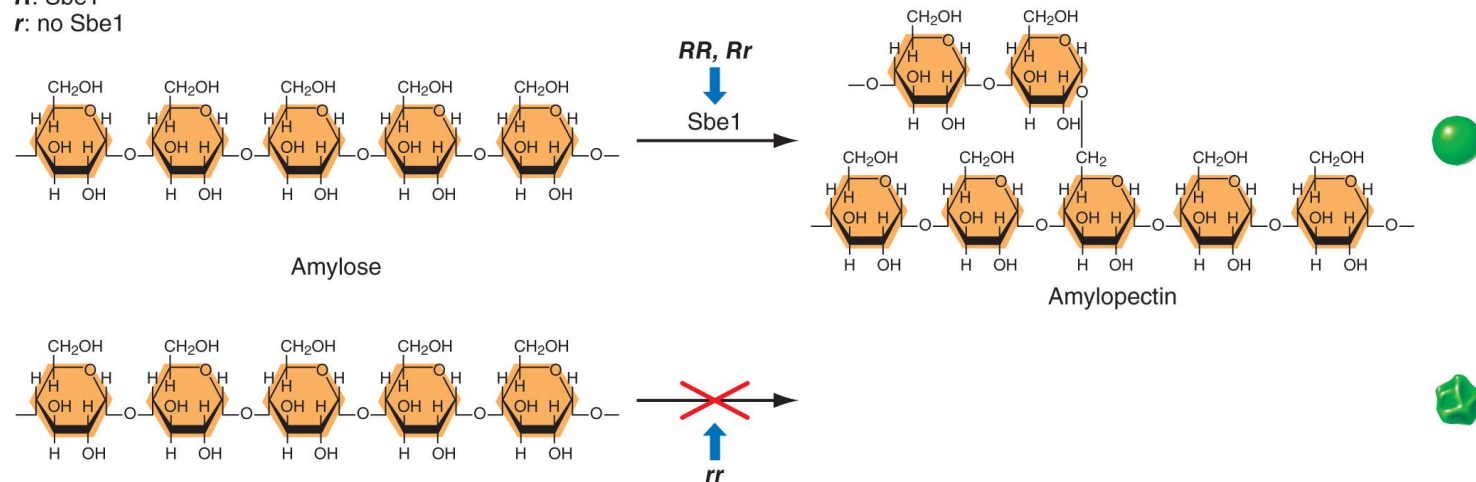
The r allele does not make Sbe1. Sucrose and unbranched starch build up, leading to wrinkled peas.

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(a) Biochemical function of Mendel's pea shape gene

R: Sbe1

r: no Sbe1



Molecular explanation for Mendel's pea color

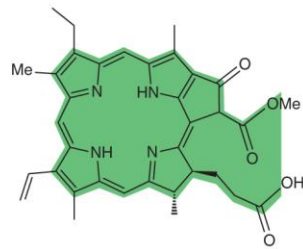
The Y allele encodes the Stay green enzyme (Sgr). Sgr helps break down chlorophyll.

The y allele does not make Sgr. Chlorophyll is not broken down and the peas remain green.

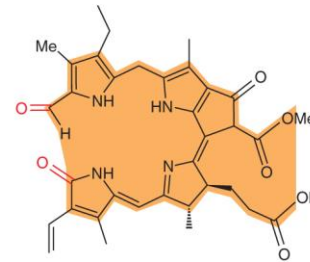
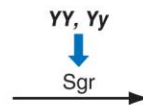
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(b) Biochemical function of Mendel's pea color gene

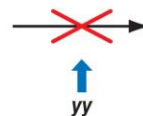
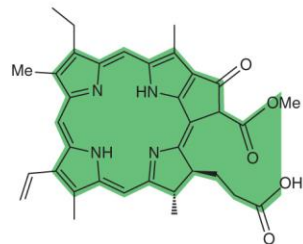
Y: Sgr
y: no Sgr



Chlorophyll (green pigment)



Breakdown product (colorless)



Two general molecular principles

- **A specific gene determines a specific enzyme, whose activity may affect phenotype.**
- **A dominant allele usually determines a normally functioning protein.**

A recessive allele usually does not encode a functional protein.

2.3 Mendelian Inheritance in Humans

Learning Objectives:

- **Analyze human pedigrees to determine if a genetic disease exhibits recessive or dominant inheritance.**
- **Describe the most common molecular mechanism for recessively inherited human genetic diseases.**
- **Explain why Huntington's disease is caused by a dominant allele.**

Mendelian inheritance in humans

Many heritable traits in humans are caused by interaction of multiple genes and so don't show simple Mendelian inheritance patterns

In 2013, there were ~ 6000 single-gene traits known in humans

Even with single-gene traits, determining inheritance pattern in humans can be tricky

- **Long generation time**
- **Small numbers of progeny**
- **No controlled matings**
- **No pure-breeding lines**

Some of the most common single-gene traits caused by recessive alleles in humans

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TABLE 2.1 Some of the Most Common Single-Gene Traits in Humans

Disease	Effect	Incidence of Disease
<i>Caused by a Recessive Allele</i>		
Thalassemia (chromosome 16 or 11)	Reduced amounts of hemoglobin; anemia, bone and spleen enlargement	1/10 in parts of Italy
Sickle-cell anemia (chromosome 11)	Abnormal hemoglobin; sickle-shaped red cells, anemia, blocked circulation; increased resistance to malaria	1/625 African-Americans
Cystic fibrosis (chromosome 7)	Defective cell membrane protein; excessive mucus production; digestive and respiratory failure	1/2000 Caucasians
Tay-Sachs disease (chromosome 15)	Missing enzyme; buildup of fatty deposit in brain; buildup disrupts mental development	1/3000 Eastern European Jews
Phenylketonuria (PKU) (chromosome 12)	Missing enzyme; mental deficiency	1/10,000 Caucasians

Some of the most common single-gene traits caused by dominant alleles in humans

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<i>Caused by a Dominant Allele</i>		
Hypercholesterolemia (chromosome 19)	Missing protein that removes cholesterol from the blood; heart attack by age 50	1/122 French Canadians
Huntington disease (chromosome 4)	Abnormal Huntingtin protein; progressive mental and neurological damage; neurologic disorders by ages 40–70	1/25,000 Caucasians

In humans, pedigrees can be used to study inheritance

Pedigrees are orderly diagrams of a family's relevant genetic features

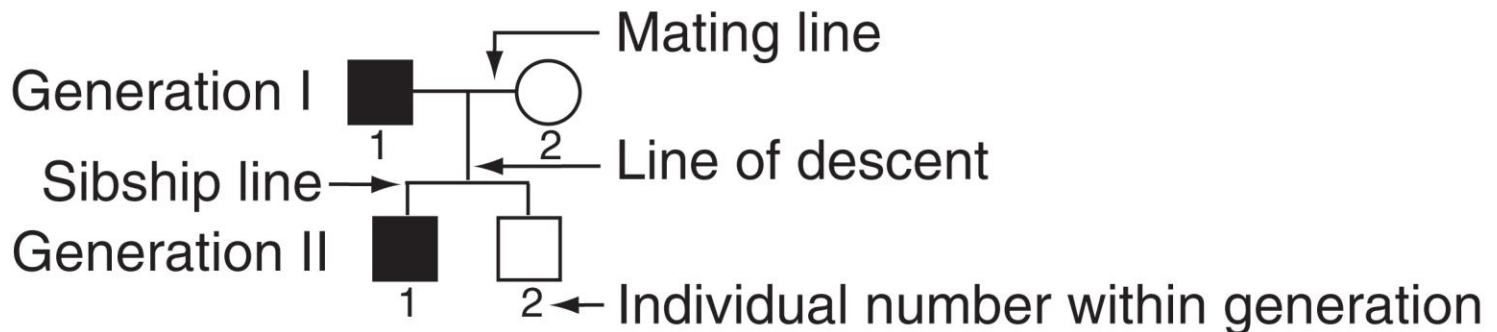
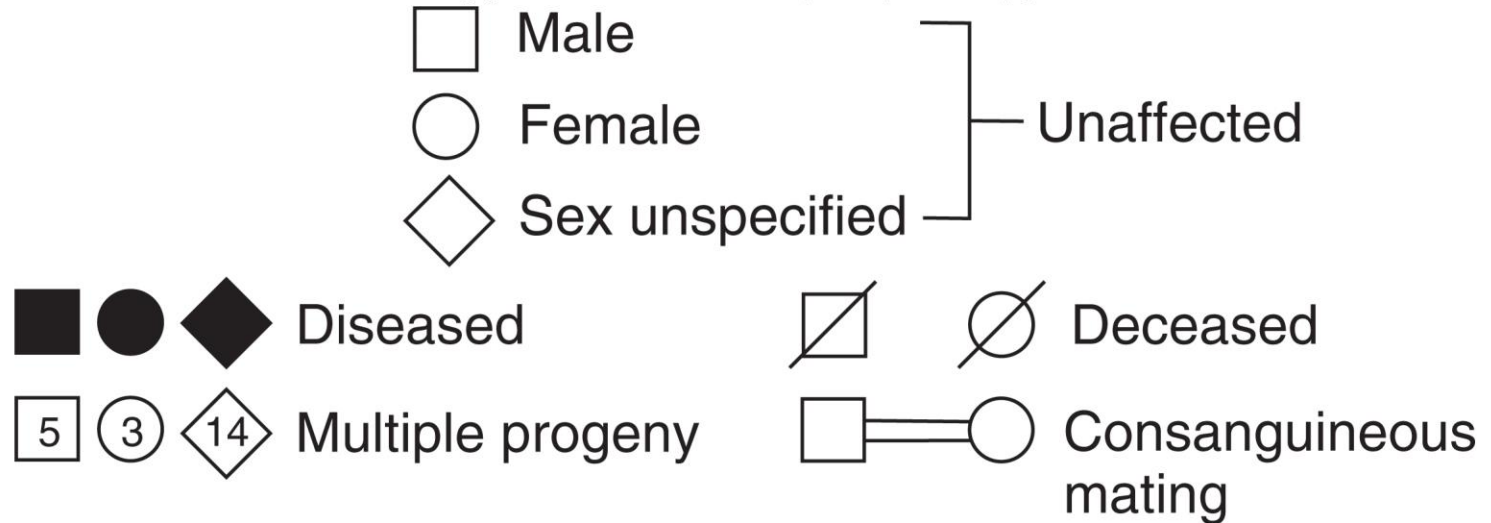
Includes as many generations as possible (ideally, at least both sets of grandparents of an affected person)

Pedigrees can be analyzed using Mendel's laws

- Is a trait determined by alternate alleles of a single gene?**
- Is a trait dominant or recessive?**

Symbols used in pedigree analysis

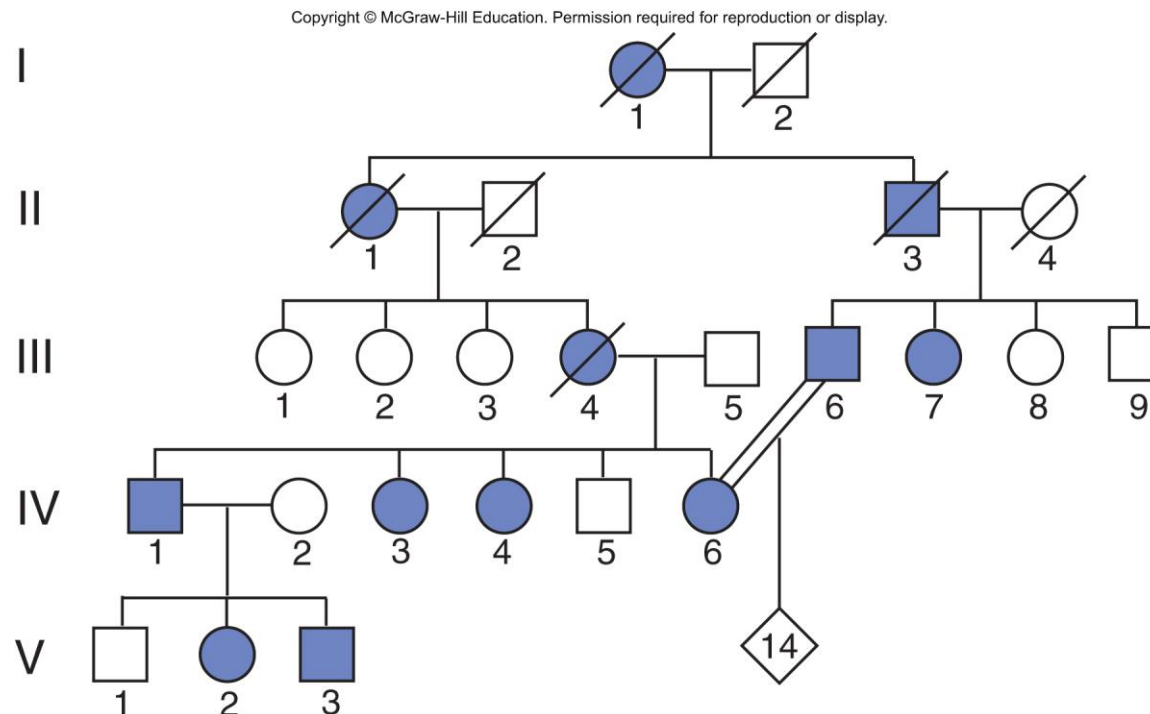
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A vertical pattern of inheritance indicates a rare dominant trait; e.g Huntington disease

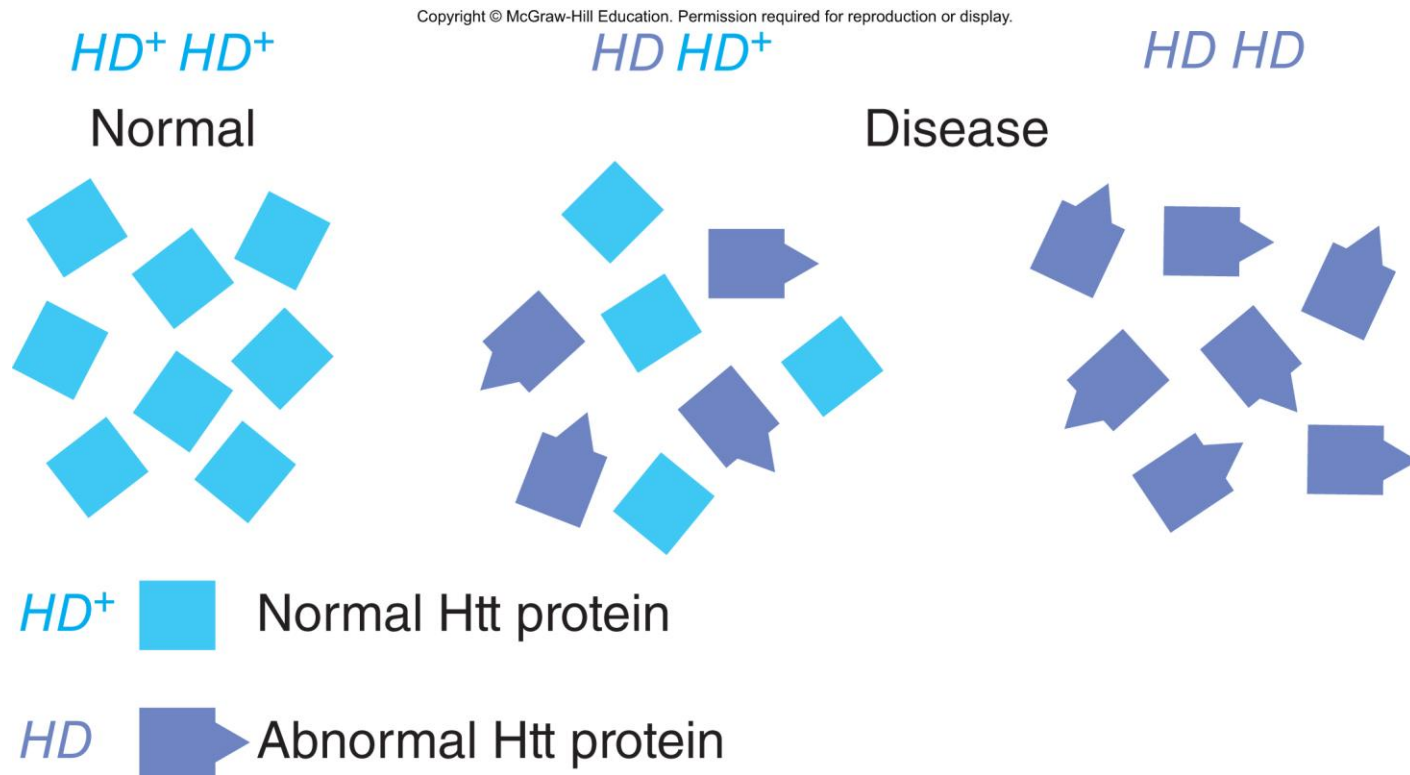
Every affected person has at least one affected parent

Mating between affected person and unaffected person is effectively a testcross



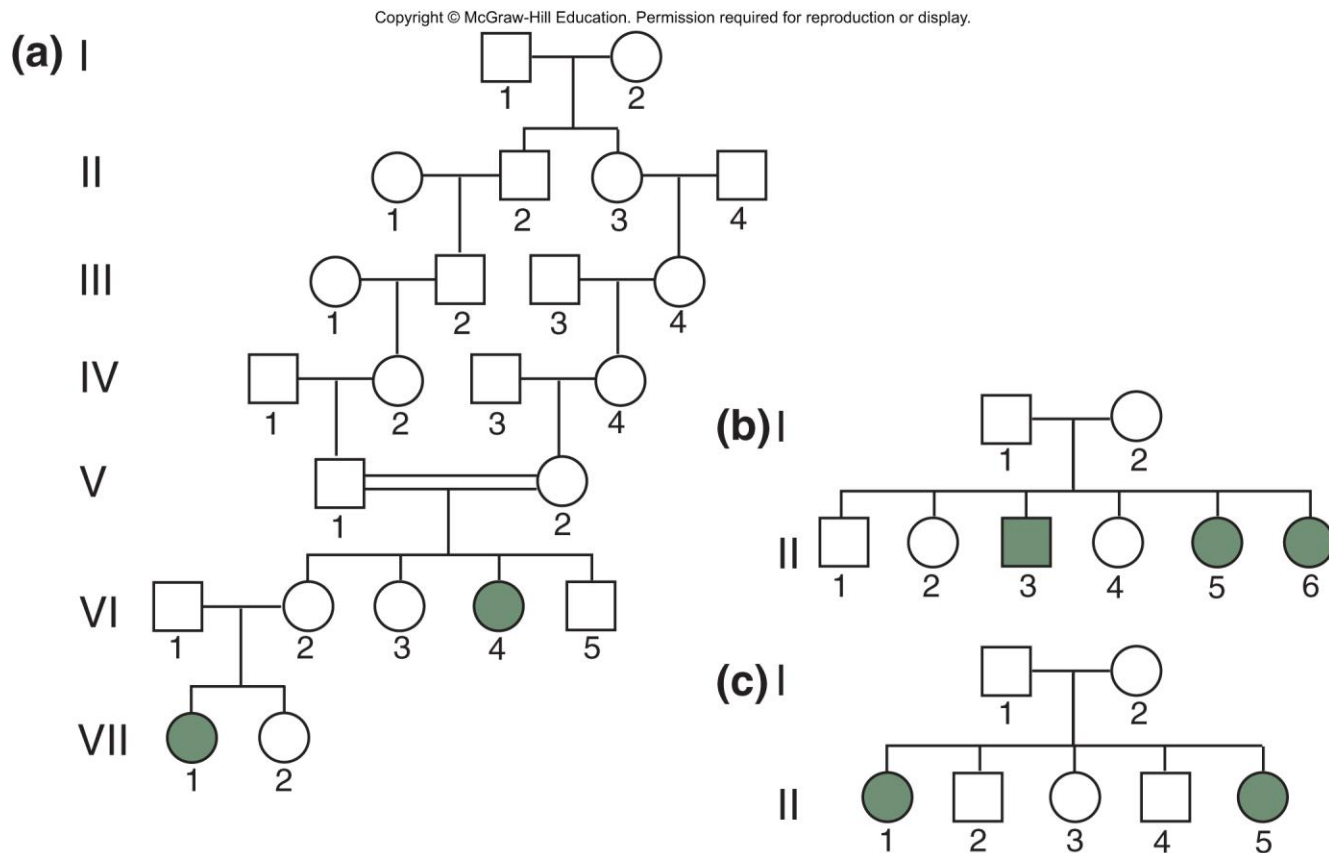
The dominant Huntington disease allele produces abnormal Htt protein

The abnormal protein damages nerve cells, even when the normal protein is also present.



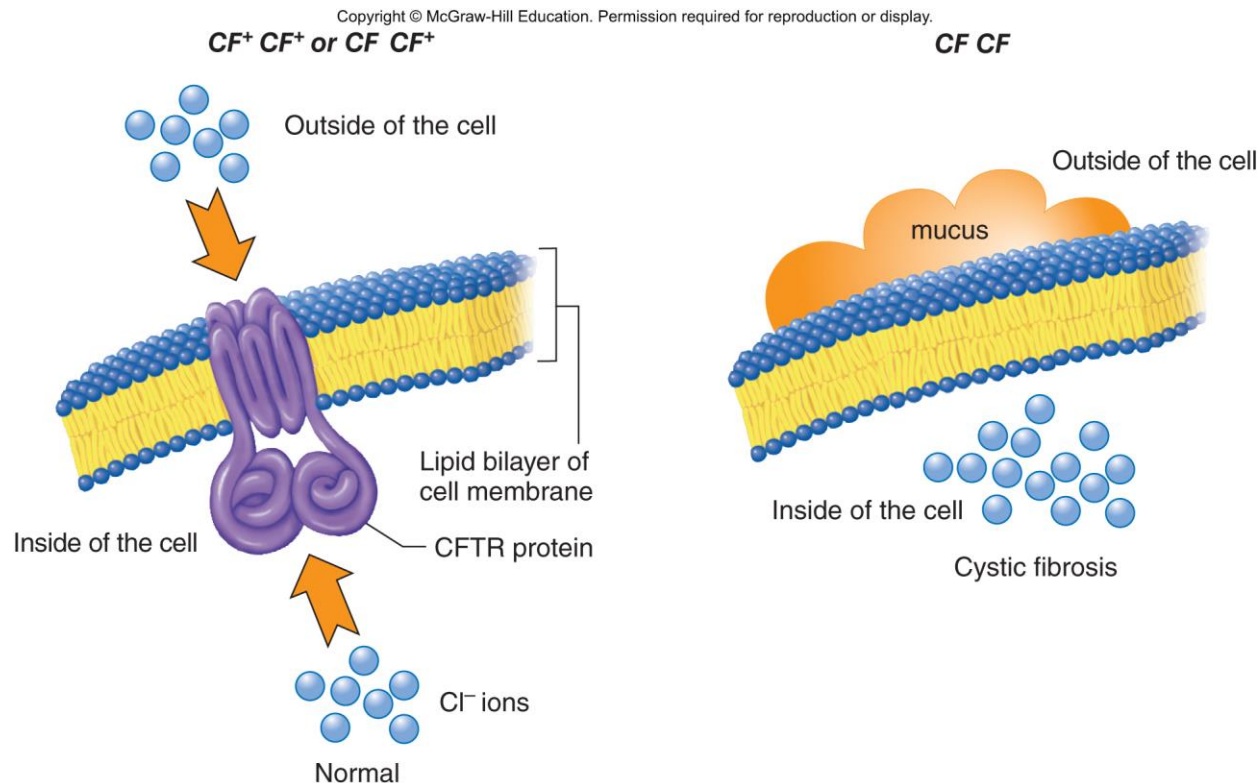
A horizontal pattern of inheritance indicates a rare recessive trait; e.g. cystic fibrosis

Parents of affected individuals are but are heterozygous (carriers) for the recessive allele



The recessive cystic fibrosis disease allele encode abnormal CFTR protein

CFTR protein regulates the passage of chloride ions across the cell membrane. Heterozygous individuals produce enough CFTR for normal lung function.



How to recognize recessive traits in pedigrees

Four keys aspects of pedigrees with recessive traits:

1. Affected individuals can be the children of two unaffected carriers, particularly as a result of consanguineous matings
2. All the children of two affected parents should be affected
3. Rare recessive traits show a **horizontal pattern** of inheritance
4. Recessive traits may show a **vertical pattern** of inheritance if the trait is extremely common in the population

Table 2.2

How to recognize dominant traits in pedigrees

Three key aspects of pedigrees with dominant traits:

- 1. Affected children always have at least one affected parent**
- 2. As a result, dominant traits show a **vertical pattern** of inheritance**
- 3. Two affected parents can produce unaffected children, if both parents are heterozygotes**

Table 2.2